Program:

Function coding:

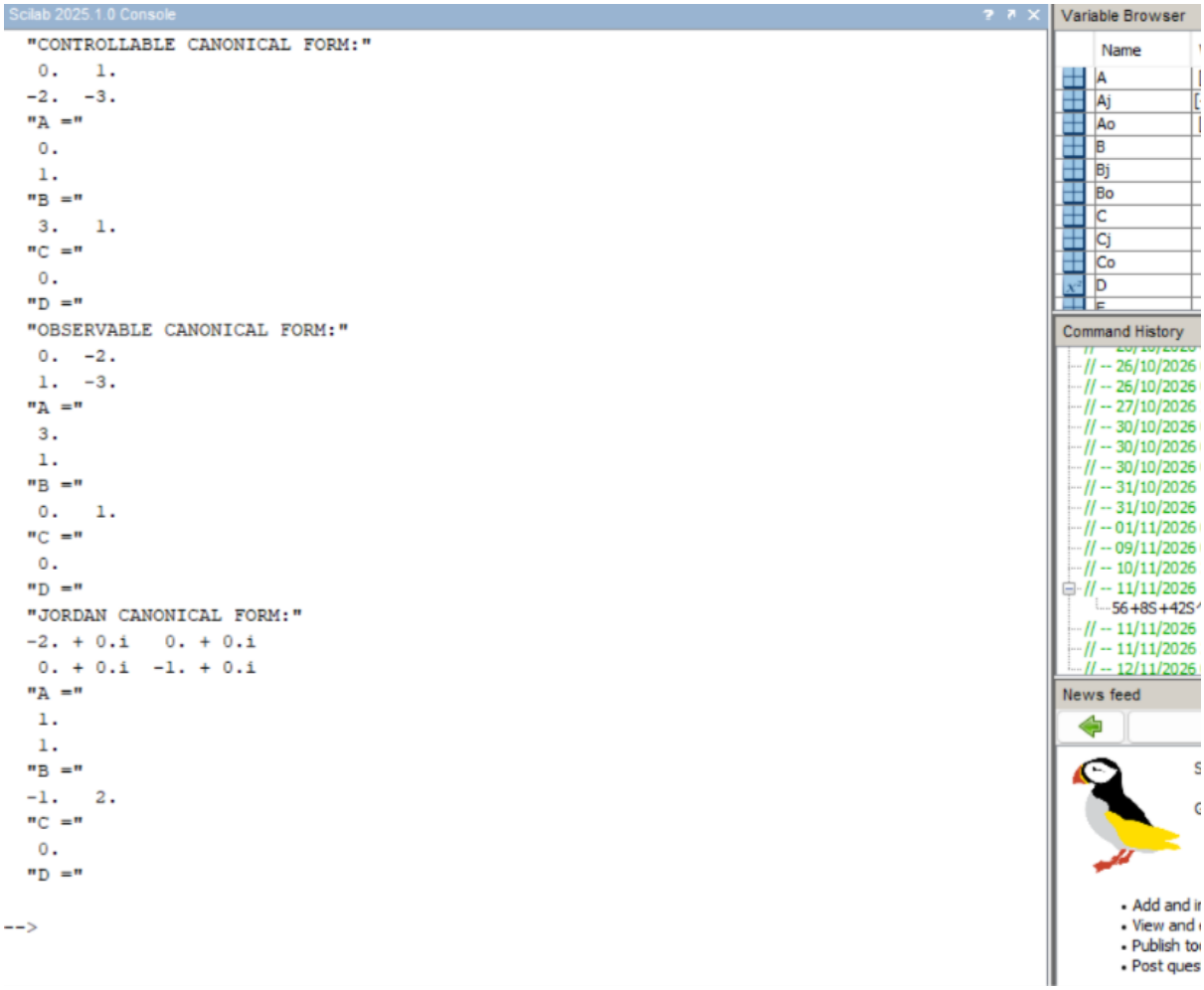
```
function [r, z, p]=pf_residue(N, D)
z=roots(N) //zeros
p=roots(D) //poles
q=round(p);
m=1; //to keep a count of the roots multiplicity
for i=1:length(p)
if(i<length(p)&q(i+1)==q(i))
m=m+1;
else
P1=N/pdiv(D,(s-p(i))^m);
r(i)=horner(P1,p(i));
for j=1:(m-1)
P1=derivat(P1);
r(i-j)=horner(P1/gamma(j+1),p(i));
end //gamma(j+1)=j! (factorial)
m=1;
end
end
endfunction
//save the file as pf_residue.sci
```

Main program:

```
clc;
clear;
xdel(winsid());
cd"C:\Users\admin\Documents\Scilab"
```

```
exec("pf_residue.sci")
s=%s;
N=s+3;
D=s^2+3*s+2;
Hc=cont_frm(N,D);
disp('Controllable form =');
ssprint(Hc);
Ho=syslin('c',(Hc.A)',(Hc.C)',(Hc.B)',Hc.D);
disp('Observable form =');
ssprint(Ho);
A=diag(roots(D));
B=[1;1];
C=pf_residue(N,D)';
D=Hc.D;
Hj=syslin('c',A,B,C,D);
disp('Jordan canonical form =');
ssprint(Hj);
```

OUTPUT:


Scilab 2025.1.0 Console
"CONTROLLABLE CANONICAL FORM:"
0. 1.
-2. -3.
"A ="
0.
1.
"B ="
3. 1.
"C ="
0.
"D ="
"OBSERVABLE CANONICAL FORM:"
0. -2.
1. -3.
"A ="
3.
1.
"B ="
0. 1.
"C ="
0.
"D ="
"JORDAN CANONICAL FORM:"
-2. + 0.i 0. + 0.i
0. + 0.i -1. + 0.i
"A ="
1.
1.
"B ="
-1. 2.
"C ="
0.
"D ="
-->
Variable Browser
Name
A
Aj
Ao
B
Bj
Bo
C
Cj
Co
D
Command History
// -- 26/10/2026
// -- 26/10/2026
// -- 27/10/2026
// -- 30/10/2026
// -- 30/10/2026
// -- 30/10/2026
// -- 31/10/2026
// -- 31/10/2026
// -- 01/11/2026
// -- 09/11/2026
// -- 10/11/2026
// -- 11/11/2026
// -- 11/11/2026
// -- 11/11/2026
News feed